

Xianfeng Wu

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RESEARCH INTERESTS

I work broadly on *generative models*, with a focus on efficient, controllable generation, and multimodal generation and understanding. My current interests center on:

- *Efficient and Controllable Generative Models*, including diffusion llms, text-to-image generation, preference alignment, knowledge distillation, and controllable image relighting.
- *Multimodal Generative Models*, including multimodal diffusion llms, 4D language-visual geometry generation and generative modeling for scientific domains such as RNA inverse folding.

EDUCATION

The University of North Carolina at Chapel Hill

Incoming Ph.D. Student in Computer Science

Aug. 2026 –

Chapel Hill, NC, USA

- Advisor: [Tianlong Chen](#)

Hong Kong University of Science and Technology

Ph.D. Student in Individualized Interdisciplinary Program

Sep. 2024 – Jun. 2025

Clear Water Bay, Hong Kong

- Advisor: [Harry Yang](#)

EXPERIENCE

Remote Research Intern

Department of Computer Science, University of North Carolina at Chapel Hill

Aug. 2025 – present

NC, USA

- Advisor: [Tianlong Chen](#)

Research Intern

School of EIC, Huazhong University of Science and Technology

Jun. 2025 – Sep. 2025

Hubei, China

- Advisors: [Xinggang Wang](#) and [Wenyu Liu](#)

Remote Research Intern

Department of Computer Science and Engineering, University at Buffalo

Jan. 2023 – Jun. 2024

NY, USA

- Advisors: [Junsong Yuan](#) and [Tianyu Luan](#)

SELECTED PUBLICATIONS

The selected publications are listed below. A full publication list can be found on my [Google Scholar](#).

(* = Equal Contribution)

#Part 1: Efficient and Controllable Generative Models

- Yuhang Chen, **Xianfeng Wu**, Jinhao Duan, Mingfu Liang, Xiaohan Wei, Yunchen Pu, Fei Tian, Chonglin Sun, Parish Aggarwal, Frank Shyu, Luke Simon, Sandeep Pandey, Xi Liu, Tianlong Chen, Bifocal Diffusion Language Models: Asymmetric Bidirectional Context for Parallel Generation, Under review. [Paper]
- Xuran Ma, Yexin Liu, Yaofu Liu, **Xianfeng Wu**, et al., “Model Reveals What to Cache: Profiling-Based Feature Reuse for Video Diffusion Models,” *ICCV*, 2025. [Code]
- **Xianfeng Wu***, Yajing Bai*, Haoze Zheng*, Harold Haodong Chen*, Yexin Liu*, et al., “LightGen: Efficient Image Generation through Knowledge Distillation and Direct Preference Optimization,” *arXiv preprint*, 2025. [Code] [Star: 130+]

#Part 2: Multimodal and 3D/4D Generative Models

- **Xianfeng Wu**, Yuhang Chen, Jie Peng, Jinhao Duan, Tianlong Chen, OmniDenoise: Diffusion MultiModal LLMs for Omni-Modal Understanding and Prediction, Under review.
- Haoze Zheng*, Zihao Wang*, **Xianfeng Wu***, Yajing Bai, Yexin Liu, Yun Li, Xiaogang Xu, Harry Yang, “Learning Latent Proxies for Controllable Single-Image Relighting,” *CVPR*, 2026.
- **Xianfeng Wu***, Yajing Bai*, Minghan Li, Xianzu Wu, Xueqi Zhao, Zhongyuan Lai, Wenyu Liu, Xinggang Wang, “4DLangVGGT: 4D Language Visual Geometry Grounded Transformer,” *arXiv preprint*, 2025. [Project] [Code] [Star: 80+]
- Yexin Liu*, Zhengyang Lian*, Yuezhe Wang*, **Xianfeng Wu***, et al., “Unveiling the Ignorance of MLLMs: Seeing Clearly, Answering Incorrectly,” *CVPR*, 2025. [Code]
- Xianzu Wu*, **Xianfeng Wu***, Tianyu Luan, Yajing Bai, Zhongyuan Lai, Junsong Yuan, “FSC: Few-point Shape Completion,” *CVPR*, 2024. [Code]

AWARDS

Gold Reviewer ICML2026

SERVICE

Conference Reviewer: ICML(2026-), ICLR(2025-), ICCV(2025-), NeurIPS(2026-)

Journal Reviewer: IEEE Transactions on Visualization and Computer Graphics (TVCG)

SKILLS

Languages: Python, MATLAB, C/C++, Java

Frameworks: PyTorch, TensorFlow

Tools: Docker, Git, Linux, Slurm